

Learning semiconductor process logic

A from-scratch, step-token GPT predicts next steps and completions; a rule validator handles anomalies with explicit process-rule attribution.

0.722

OOD Top-1

train 12 / test 3

0.999

OOD Top-5

family-wise avg

0.263

Task 2 NED

lower is better

1.00

Task 3 F1

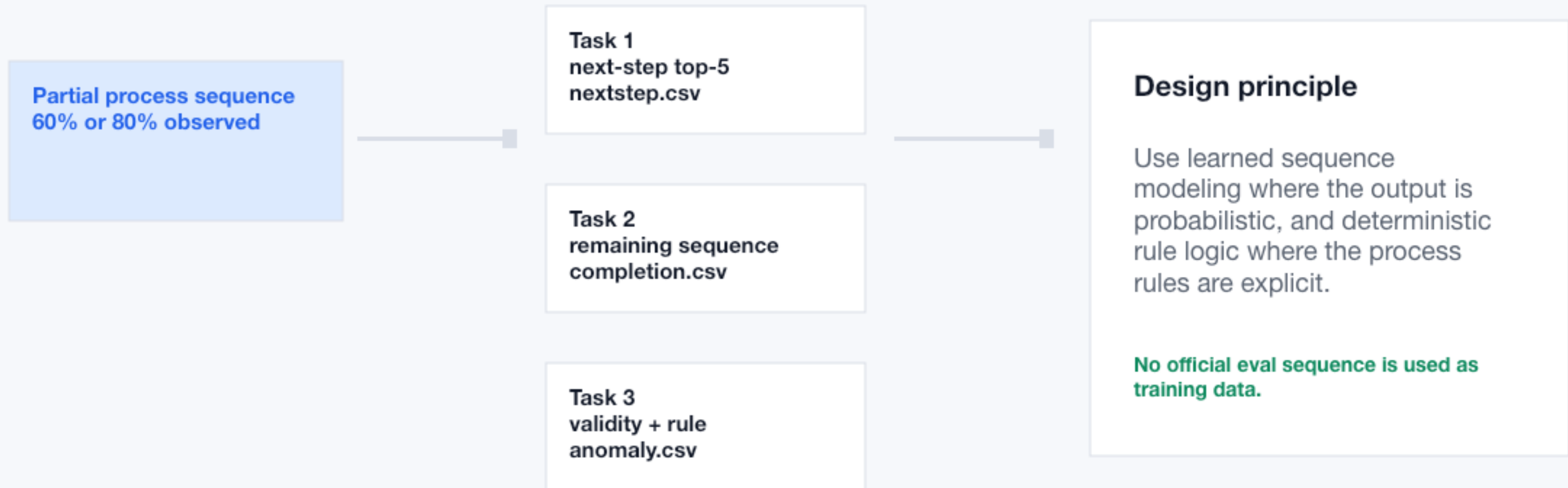
rule validator

Final package

- nextstep.csv
- completion.csv
- anomaly.csv
- trained checkpoint
- score reports

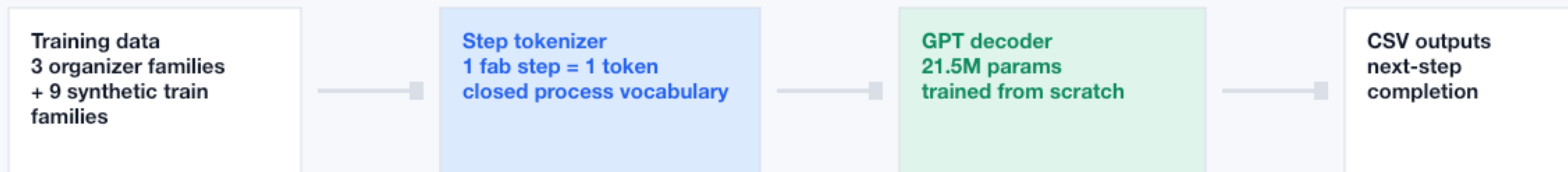
The challenge is three outputs from one process prefix

The jury scores files, but the engineering problem is process grammar generalization.



What we built: a small process model, not a generic LLM wrapper

Each process step is one token, so training optimizes exactly the scored unit.



Anomaly path: organizer process rules -> deterministic validator -> validity, score, and official rule name

The n-gram baseline remains in the repo as a dependency-light fallback and an honest comparator.

What makes a process family different?

Same vocabulary, different valid process routes. That is what makes the OOD test meaningful.

Litho levels

Number of mask / photoresist patterning cycles. More levels means a deeper route.

Extra via block

Some families repeat field oxide, via etch, via fill, and CMP.

Spacer rate

Probability that spacer dielectric and anisotropic spacer etch appear.

Implant route

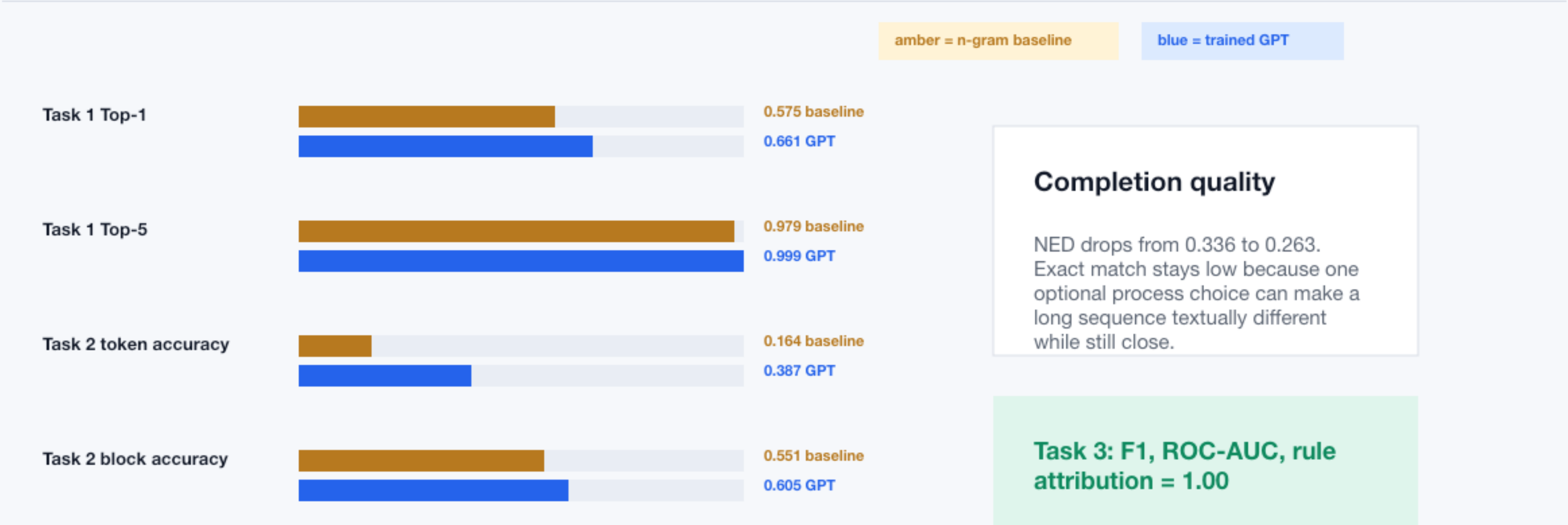
Valid recombinations such as source/drain followed by N-buffer.

Group	Litho levels	Extra via	Spacer rate	Role	Why it matters
Organizer	4 / 6 / 4	native	native	train	Real known process families
SCFAM01-09	4-6	mixed	0.25-0.80	train	Teaches varied valid routes
SCFAM10-12	4-6	mixed	0.20-0.90	unseen test	Tests hidden-family behavior

OOD claim: we keep the step vocabulary fixed, then vary structure, optional blocks, and route depth. The model must learn process order instead of memorizing one recipe.

The trained model improves the scored tasks

Organizer scorer on our self-eval split, plus the frozen 15-family OOD benchmark.



Same prefix, different model behavior

A real self-eval MOSFET prefix at the 80% cut.

Prefix tail: DEPOSIT PASSIVATION | CURE PASSIVATION | MEASURE PASSIVATION QUALITY | OPEN PAD WINDOW | OPEN PAD WINDOW LITHO | DEVELOP PHOTORESIST | PASSIVATION ETCH PAD OPENING

Ground truth next step
STRIP PHOTORESIST

N-gram baseline top-5

1. STRIP RESIST
2. STRIP PHOTORESIST
3. DEVELOP PHOTORESIST
4. SOFT BAKE
5. SPIN COAT PHOTORESIST



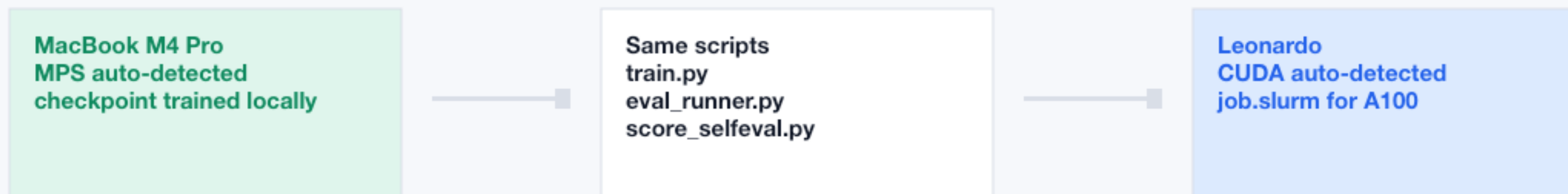
Trained GPT top-5

1. STRIP PHOTORESIST
2. STRIP RESIST
3. STRIP RESIST LEVEL 2
4. FINAL CLEAN
5. OPEN PAD WINDOW

The GPT puts the exact next step at rank 1.

Reproducible on a laptop or Leonardo

The repository contains the checkpoint, loss curve, scorer outputs, and exact commands.



Submission-readiness checklist

- Public GitHub repository
- MIT license
- REPORT.md and requirements.txt
- Official-format CSV IDs
- No official eval sequences used for training

What we would trust in production

Learned process probabilities, explicit rules, and measured OOD behavior.

0.999

Next-step

OOD Top-5

0.263

Completion

self-eval NED

1.00

Anomaly

F1 + rules

15

Families

12 train / 3 test

Kremsians built this as a full benchmark, not just a single CSV: baseline, trained model, OOD protocol, official scorer reports, checkpoint, and demo are all in the repository.

Next engineering step: validator-guided beam search for completion.